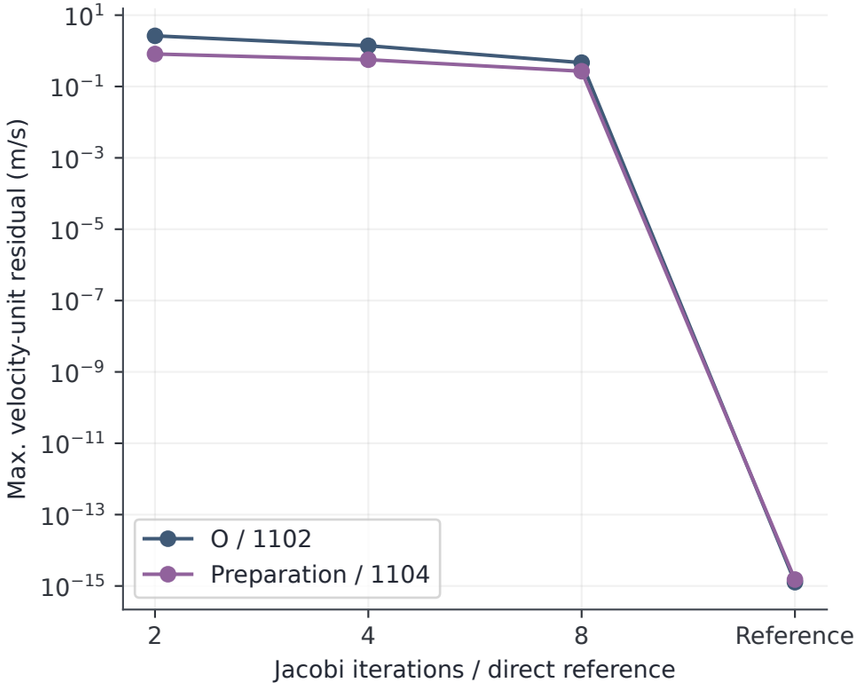
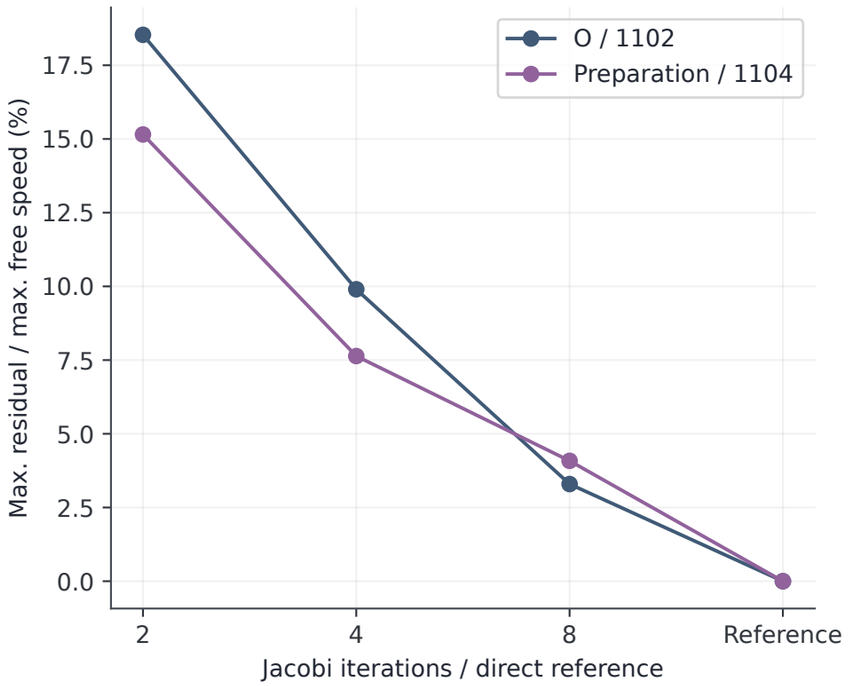
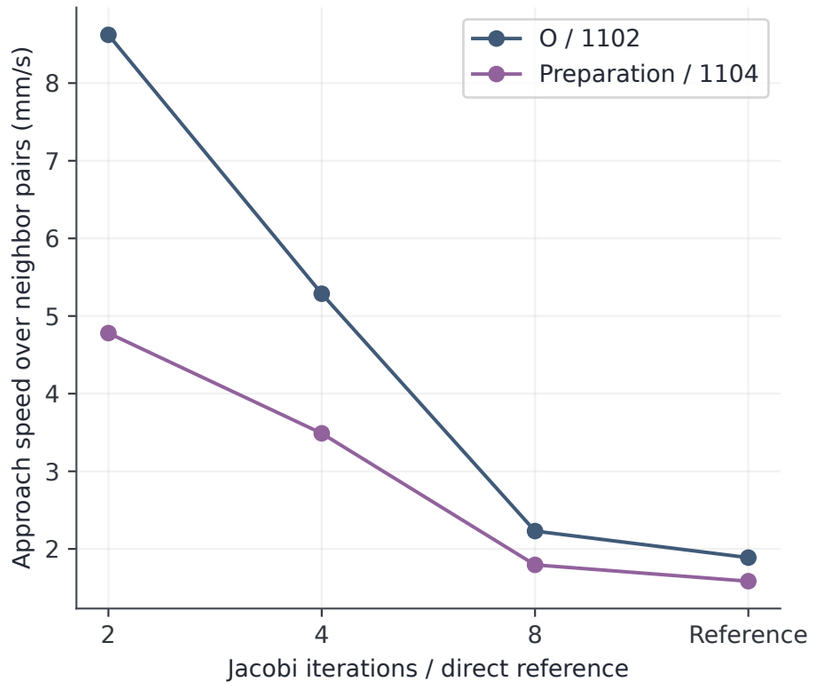
a Residual of the same resistance system



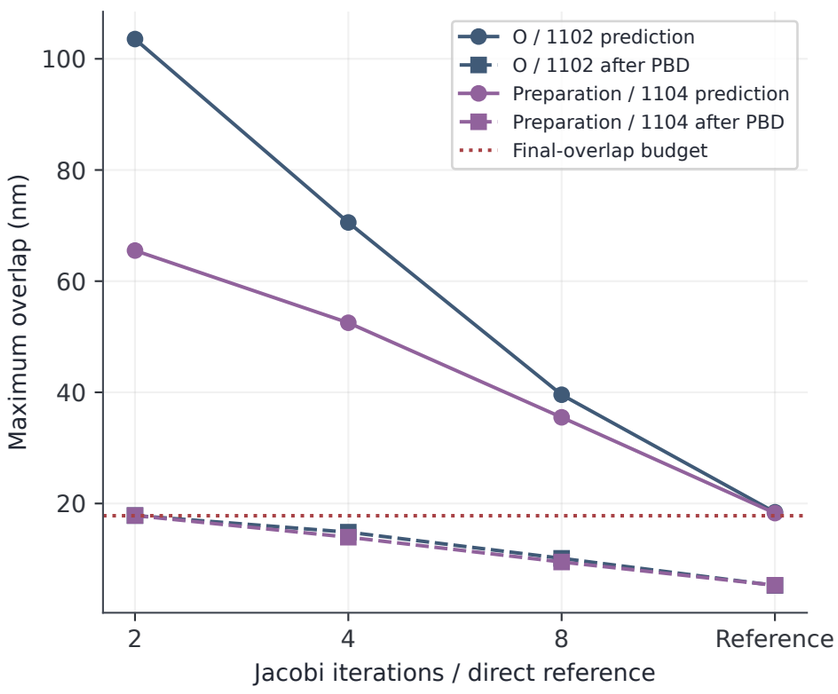
b Block-preconditioned residual



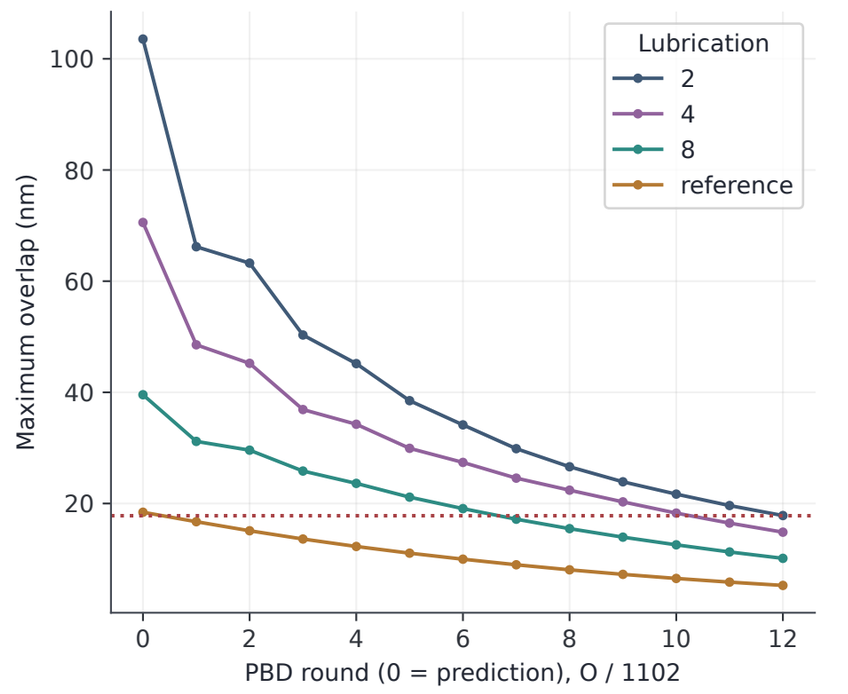
c Fastest normal approach



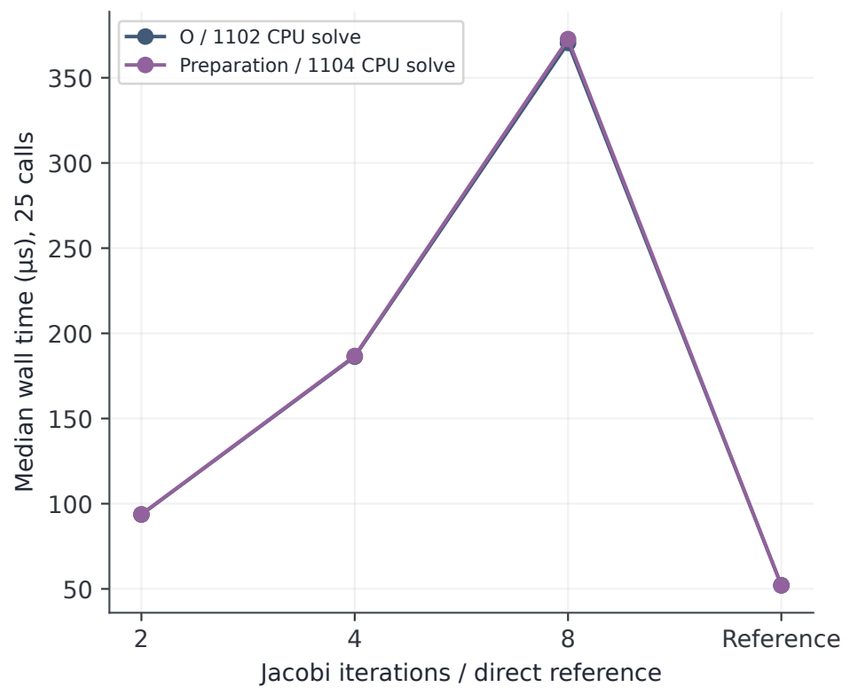
d Same state, same force, same 10 µs



e Original 12 Jacobi PBD passes



f CPU solve cost, preassembled operator



Frozen F10E first-failure inputs; original gap floor, forces, CFD and 10 µs step. Mean balance is included in panels a–d.
Algebraic convergence is not experimental hydrodynamic accuracy. CPU matrix assembly excluded in panel f; no new trajectory in this figure.